

or the philosopher possesses today a notion of 'change of meaning' applicable to such cases as the one we have been discussing. The *job* for which the notion of change of meaning was developed in the history of the language was just a *much* cruder job than this one.

But, if we don't assert either (1) or (2) – in other words, if we regard the 'change of meaning' issue as a pseudo-issue in this case – then how are we to discuss the question with which we started? 'Is pain a brain state?'

The answer is to allow statements of the form 'pain is *A*', where 'pain' and '*A*' are in no sense synonyms, and to see whether any such statement can be found which might be acceptable on empirical and methodological grounds. This is what we shall now proceed to do.

II. Is pain a brain state?

We shall discuss 'Is pain a brain state?' then. And we have agreed to waive the 'change of meaning' issue.

Since I am discussing not what the concept of pain comes to, but what pain is, in a sense of 'is' which requires empirical theory-construction (or, at least, empirical speculation), I shall not apologize for advancing an empirical hypothesis. Indeed, my strategy will be to argue that pain is *not* a brain state, not on *a priori* grounds, but on the grounds that another hypothesis is more plausible. The detailed development and verification of my hypothesis would be just as Utopian a task as the detailed development and verification of the brain-state hypothesis. But the putting-forward, not of detailed and scientifically 'finished' hypotheses, but of schemata for hypotheses, has long been a function of philosophy. I shall, in short, argue that pain is not a brain state, in the sense of a physical-chemical state of the brain (or even the whole nervous system), but another *kind* of state entirely. I propose the hypothesis that pain, or the state of being in pain, is a functional state of a whole organism.

To explain this it is necessary to introduce some technical notions. In previous papers I have explained the notion of a Turing Machine and discussed the use of this notion as a model for an organism. The notion of a Probabilistic Automaton is defined similarly to a Turing Machine, except that the transitions between 'states' are allowed to be with various probabilities rather than being 'deterministic'. (Of course, a Turing Machine is simply a special kind of Probabilistic Automaton, one with transition probabilities 0, 1). I shall assume the notion of a Probabilistic Automaton has been generalized to allow for 'sensory inputs' and 'motor outputs' – that is, the Machine Table specifies, for

every possible combination of a 'state' and a complete set of 'sensory inputs', an 'instruction' which determines the probability of the next 'state', and also the probabilities of the 'motor outputs'. (This replaces the idea of the Machine as printing on a tape.) I shall also assume that the physical realization of the sense organs responsible for the various inputs, and of the motor organs, is specified, but that the 'states' and the 'inputs' themselves are, as usual, specified only 'implicitly' – i.e. by the set of transition probabilities given by the Machine Table.

Since an empirically given system can simultaneously be a 'physical realization' of many different Probabilistic Automata, I introduce the notion of a *Description* of a system. A Description of S where S is a system, is any true statement to the effect that S possesses distinct states $S_1, S_2 \dots S_n$ which are related to one another and to the motor outputs and sensory inputs by the transition probabilities given in such-and-such a Machine Table. The Machine Table mentioned in the Description will then be called the Functional Organization of S relative to that Description, and the S_i such that S is in state S_i at a given time will be called the Total State of S (at the time) relative to that Description. It should be noted that knowing the Total State of a system relative to a Description involves knowing a good deal about how the system is likely to 'behave', given various combinations of sensory inputs, but does *not* involve knowing the physical realization of the S_i as, e.g. physical-chemical states of the brain. The S_i , to repeat, are specified only *implicitly* by the Description – i.e. specified *only* by the set of transition probabilities given in the Machine Table.

The hypothesis that 'being in pain is a functional state of the organism' may now be spelled out more exactly as follows:

- (1) All organisms capable of feeling pain are Probabilistic Automata.
- (2) Every organism capable of feeling pain possesses at least one Description of a certain kind (i.e. being capable of feeling pain *is* possessing an appropriate kind of Functional Organization).
- (3) No organism capable of feeling pain possesses a decomposition into parts which separately possess Descriptions of the kind referred to in (2).
- (4) For every Description of the kind referred to in (2), there exists a subset of the sensory inputs such that an organism with that Description is in pain when and only when some of its sensory inputs are in that subset.

This hypothesis is admittedly vague, though surely no vaguer than the brain-state hypothesis in its present form. For example, one would like to know more about the kind of Functional Organization that an organism must have to be capable of feeling pain, and more about the

marks that distinguish the subset of the sensory inputs referred to in (4). With respect to the first question, one can probably say that the Functional Organization must include something that resembles a 'preference function', or at least a preference partial ordering and something that resembles an 'inductive logic' (i.e. the Machine must be able to 'learn from experience'). (The meaning of these conditions, for Automata models, is discussed in the previous chapter.) In addition, it seems natural to require that the Machine possess 'pain sensors', i.e. sensory organs which normally signal damage to the Machine's body, or dangerous temperatures, pressures, etc., which transmit a special subset of the inputs, the subset referred to in (4). Finally, and with respect to the second question, we would want to require at least that the inputs in the distinguished subset have a high disvalue on the Machine's preference function or ordering (further conditions are discussed in the previous chapter). The purpose of condition (3) is to rule out such 'organisms' (if they can count as such) as swarms of bees as single pain-feelers. The condition (1) is, obviously, redundant, and is only introduced for expository reasons. (It is, in fact, empty, since everything is a Probabilistic Automaton under *some* Description.)

I contend, in passing, that this hypothesis, in spite of its admitted vagueness, is far *less* vague than the 'physical-chemical state' hypothesis is today, and far more susceptible to investigation of both a mathematical and an empirical kind. Indeed, to investigate this hypothesis is just to attempt to produce 'mechanical' models of organisms – and isn't this, in a sense, just what psychology is about? The difficult step, of course, will be to pass from models to *specific* organisms to a *normal form* for the psychological description of organisms – for this is what is required to make (2) and (4) precise. But this too seems to be an inevitable part of the program of psychology.

I shall now compare the hypothesis just advanced with (a) the hypothesis that pain is a brain state, and (b) the hypothesis that pain is a behavior disposition.

III. Functional state versus brain state

It may, perhaps, be asked if I am not somewhat unfair in taking the brain-state theorist to be talking about *physical-chemical* states of the brain. But (a) these are the only sorts of states ever mentioned by brain-state theorists. (b) The brain-state theorist usually mentions (with a certain pride, slightly reminiscent of the Village Atheist) the incompatibility of his hypothesis with all forms of dualism and mentalism. This is natural if physical-chemical states of the brain are what is at

issue. However, functional states of whole systems are something quite different. In particular, the functional-state hypothesis is *not* incompatible with dualism! Although it goes without saying that the hypothesis is 'mechanistic' in its inspiration, it is a slightly remarkable fact that a system consisting of a body and a 'soul', if such things there be, can perfectly well be a Probabilistic Automaton. (c) One argument advanced by Smart is that the brain-state theory assumes only 'physical' properties, and Smart finds 'non-physical' properties unintelligible. The Total States and the 'inputs' defined above are, of course, neither mental nor physical *per se*, and I cannot imagine a functionalist advancing this argument. (d) If the brain-state theorist does mean (or at least allow) states other than physical-chemical states, then his hypothesis is completely empty, at least until he specifies *what* sort of 'states' he *does* mean.

Taking the brain-state hypothesis in this way, then, what reasons are there to prefer the functional-state hypothesis over the brain-state hypothesis? Consider what the brain-state theorist has to do to make good his claims. He has to specify a physical-chemical state such that *any* organism (not just a mammal) is in pain if and only if (a) it possesses a brain of a suitable physical-chemical structure; and (b) its brain is in that physical-chemical state. This means that the physical-chemical state in question must be a possible state of a mammalian brain, a reptilian brain, a mollusc's brain (octopuses are mollusca, and certainly feel pain), etc. At the same time, it must *not* be a possible (physically possible) state of the brain of any physically possible creature that cannot feel pain. Even if such a state can be found, it must be nomologically certain that it will also be a state of the brain of any extra-terrestrial life that may be found that will be capable of feeling pain before we can even entertain the supposition that it may *be* pain.

It is not altogether impossible that such a state will be found. Even though octopus and mammal are examples of parallel (rather than sequential) evolution, for example, virtually identical structures (physically speaking) have evolved in the eye of the octopus and in the eye of the mammal, notwithstanding the fact that this organ has evolved from different kinds of cells in the two cases. Thus it is at least possible that parallel evolution, all over the universe, might *always* lead to *one and the same* physical 'correlate' of pain. But this is certainly an ambitious hypothesis.

Finally, the hypothesis becomes still more ambitious when we realize that the brain-state theorist is not just saying that *pain* is a brain state; he is, of course, concerned to maintain that *every* psychological state is a brain state. Thus if we can find even one psychological predi-

cate which can clearly be applied to both a mammal and an octopus (say 'hungry'), but whose physical-chemical 'correlate' is different in the two cases, the brain-state theory has collapsed. It seems to me overwhelmingly probable that we can do this. Granted, in such a case the brain-state theorist can save himself by *ad hoc* assumptions (e.g. defining the disjunction of two states to be a single 'physical-chemical state'), but this does not have to be taken seriously.

Turning now to the considerations *for* the functional-state theory, let us begin with the fact that we identify organisms as in pain, or hungry, or angry, or in heat, etc., on the basis of their *behavior*. But it is a truism that similarities in the behavior of two systems are at least a reason to suspect similarities in the functional organization of the two systems, and a much *weaker* reason to suspect similarities in the actual physical details. Moreover, we expect the various psychological states – at least the basic ones, such as hunger, thirst, aggression, etc. – to have more or less similar 'transition probabilities' (within wide and ill defined limits, to be sure) with each other and with behavior in the case of different species, because this is an artifact of the way in which we identify these states. Thus, we would not count an animal as *thirsty* if its 'unsatiated' behavior did not seem to be directed toward drinking and was not followed by 'satiation for liquid'. Thus any animal that we count as capable of these various states will at least *seem* to have a certain rough kind of functional organization. And, as already remarked, if the program of finding psychological laws that are not species-specific – i.e. of finding a normal form for psychological theories of different species – ever succeeds, then it will bring in its wake a delineation of the kind of functional organization that is necessary and sufficient for a given psychological state, as well as a precise definition of the notion 'psychological state'. In contrast, the brain-state theorist has to hope for the eventual development of neurophysiological laws that are species-independent, which seems much less reasonable than the hope that psychological laws (of a sufficiently general kind) may be species-independent, or, still weaker, that a species-independent *form* can be found in which psychological laws can be written.

IV. Functional state versus behavior-disposition

The theory that being in pain is neither a brain state nor a functional state but a behavior disposition has one apparent advantage: it appears to agree with the way in which we verify that organisms are in pain. We do not in practice know anything about the brain state of an animal when we say that it is in pain; and we possess little if any knowledge of